

NONG PHLUB AGROMET, Thailand

WMO: 484740

Lat: 12.5893N

Lon: 99.7346E

Elev: 354

StdP: 14.51

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.3	65.0	53.1	60.7	70.4	56.1	67.8	72.3	15.4	84.1	13.9	83.7	0.5	N/A	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.0	98.2	77.6	96.2	77.7	94.6	77.7	80.8	90.9	80.2	90.0	79.8	89.4	6.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	149.7	84.5	78.0	147.2	84.0	77.4	144.3	83.5	44.8	91.2	44.2	90.2	43.6	89.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.2	11.7	10.2	DB	58.3	101.5	3.2	2.1	56.0	103.0	54.2	104.3	52.4	105.5	50.1	107.0	
			WB	55.6	82.2	3.5	0.6	53.1	82.7	51.1	83.0	49.2	83.3	46.6	83.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	77.7	80.5	83.3	85.5	84.8	83.7	82.9	82.9	82.3	81.1	79.8	77.6
	DBStd	3.56	3.53	2.82	3.01	2.91	2.33	2.00	1.90	1.88	1.76	1.88	2.28	3.54
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11623	858	853	1033	1066	1079	1011	1021	1019	969	964	894	856
	CDD65	6148	393	433	568	616	614	561	556	554	519	499	444	391
	CDH74	60714	3377	4282	6266	7084	6597	5928	5669	5499	4790	4303	3722	3197
	CDH80	25503	1291	1977	3193	3645	2951	2399	2238	2147	1782	1445	1299	1136
Wind	WSAvg	1.9	2.4	2.3	2.4	2.1	1.6	1.8	1.6	1.5	1.3	1.3	2.1	2.6
Precipitation	PrecAvg	38.4	0.6	0.4	2.0	1.7	4.1	3.5	3.9	3.3	5.7	10.3	2.5	0.4
	PrecMax	54.5	2.7	3.0	16.6	5.2	14.0	6.5	6.8	6.9	11.7	21.6	12.4	1.6
	PrecMin	27.3	0.0	0.0	0.0	0.0	1.1	1.3	1.1	1.4	3.3	3.5	0.0	0.0
	PrecStd	6.7	0.7	0.7	3.3	1.4	2.6	1.3	1.5	1.1	1.6	5.0	2.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.5	95.4	99.4	101.6	98.9	96.1	95.1	94.9	94.2	92.0	92.3	91.8
		MCWB	74.8	75.7	75.3	77.9	79.3	78.8	78.4	78.2	78.4	78.9	77.1	75.4
	2%	DB	90.5	93.5	96.9	99.1	96.3	94.0	93.3	92.8	92.1	90.4	90.2	90.1
		MCWB	74.2	75.3	76.0	78.0	79.3	78.7	78.1	78.0	78.5	78.6	76.5	75.1
	5%	DB	88.9	91.8	95.1	96.9	94.3	92.2	91.5	91.2	90.5	89.0	88.7	88.4
		MCWB	73.5	75.1	76.1	78.0	79.3	78.6	78.0	77.7	78.3	78.3	76.4	74.4
	10%	DB	87.0	90.0	93.3	94.6	92.1	90.3	89.6	89.4	88.6	87.4	87.1	86.4
		MCWB	73.0	74.8	76.1	78.0	79.2	78.5	77.8	77.5	78.0	78.1	76.1	73.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.2	78.5	79.7	81.3	81.8	81.2	80.4	80.1	80.7	80.6	79.7	78.2
		MCDB	86.8	90.4	89.9	94.3	92.2	90.2	90.0	89.0	89.1	88.5	86.9	86.9
	2%	WB	76.9	77.4	78.6	80.4	80.9	80.3	79.5	79.2	79.8	79.7	78.7	77.0
		MCDB	86.1	88.9	90.2	92.8	91.2	89.4	89.1	88.2	88.2	87.3	86.2	86.1
	5%	WB	76.0	76.7	77.9	79.7	80.1	79.7	78.9	78.7	79.1	79.0	78.0	76.0
		MCDB	84.5	87.8	89.5	91.8	90.2	88.7	88.4	87.7	87.4	86.5	85.2	84.7
	10%	WB	75.1	76.0	77.3	78.9	79.6	79.0	78.2	78.1	78.4	78.5	77.3	75.1
		MCDB	83.0	86.7	88.8	90.8	89.3	87.8	87.1	86.7	86.3	85.4	84.1	83.2
Mean Daily Temperature Range	5% DB	MDBR	19.1	20.6	21.3	20.0	16.6	14.4	14.2	14.6	14.4	13.0	13.9	17.0
		MCDBR	21.3	22.6	23.9	23.2	19.5	17.3	17.2	17.4	17.1	15.5	17.2	19.3
		MCWBR	7.0	6.9	6.7	5.9	5.4	5.0	5.1	4.9	5.5	5.4	5.8	6.3
	5% WB	MCDBR	16.9	19.8	21.6	20.9	17.4	15.2	15.2	15.4	15.6	13.9	14.2	16.7
	MCWBR	6.0	6.6	6.8	5.9	5.5	5.2	5.2	5.1	5.7	5.4	5.4	6.2	
Clear-Sky Solar Irradiance	taub	0.512	0.555	0.571	0.518	0.458	0.438	0.436	0.437	0.440	0.518	0.506	0.493	
	taud	2.058	1.965	1.962	2.149	2.359	2.430	2.429	2.415	2.394	2.132	2.136	2.133	
	Ebn at Noon	251	245	244	256	267	269	271	274	275	252	251	252	
	Edh at Noon	52	59	61	50	40	36	37	38	39	49	48	48	
All-Sky Solar Radiation	RadAvg	1554	1727	1767	1848	1597	1439	1354	1373	1421	1374	1470	1502	
	RadStd	109	89	149	125	156	114	116	124	111	111	129	96	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page